

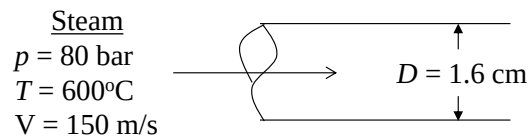
Problem 4.3

Steam enters a 1.6-cm-diameter pipe at 80 bar and 600°C with a velocity of 150 m/s. Determine the mass flow rate, in kg/s.

KNOWN: Pressure, temperature, and velocity of steam entering a 1.6-cm-diameter pipe are specified.

FIND: Determine the mass flow rate.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL:

1. Flow is one-dimensional.

ANALYSIS:

The mass flow rate for one-dimensional flow in terms of specific volume is

$$\dot{m} = \frac{AV}{v}$$

At $p = 80 \text{ bar}$ and $T = 600^\circ\text{C}$, the steam is superheated vapor. Obtaining specific volume of the steam from Table A-4: $v = 0.04845 \text{ m}^3/\text{kg}$.

Cross-sectional area of the flow based on pipe diameter is

$$A = \frac{\pi}{4} D^2 = \frac{\pi}{4} (1.6 \text{ cm})^2 \left| \frac{1 \text{ m}^2}{10000 \text{ cm}^2} \right| = 0.00020 \text{ m}^2$$

Solving for the mass flow rate yields

$$\dot{m} = \frac{(0.00020 \text{ m}^2)(150 \text{ m/s})}{0.04845 \text{ m}^3/\text{kg}} = 0.62 \text{ kg/s}$$



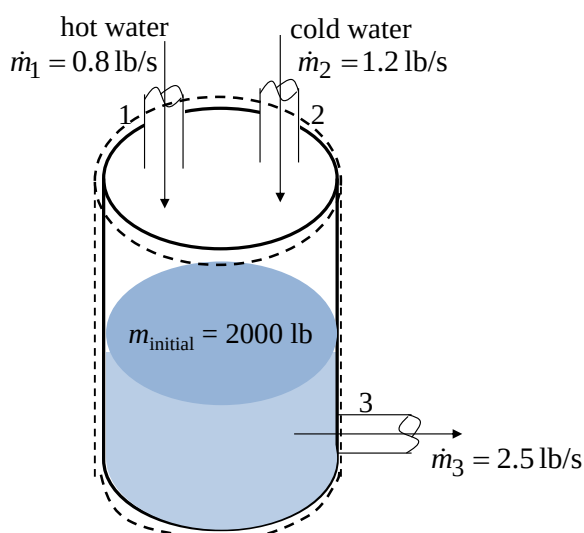
Problem 4.6

Figure P4.6 shows a mixing tank initially containing 2000 lb of liquid water. The tank is fitted with two inlet pipes, one delivering hot water at a mass flow rate of 0.8 lb/s and the other delivering cold water at a mass flow rate of 1.2 lb/s. Water exits through a single exit pipe at a mass flow rate of 2.5 lb/s. Determine the mass of water, in lb, in the tank after 30 minutes.

KNOWN: The initial mass of water in a tank, two inlet mass flow rates, and the exit mass flow rate are specified.

FIND: Determine the mass of water in the tank after 30 minutes.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL:

1. The control volume encloses the water in the tank and has two inlets and one exit.
2. The entering and exiting mass flow rates each remain constant.

ANALYSIS:

The mass rate balance for the control volume with two inlets and one exit is

$$\frac{dm_{cv}}{dt} = \dot{m}_1 + \dot{m}_2 - \dot{m}_3$$

Integrating and solving for the final mass of water in the tank

$$m_2 - m_1 = (\dot{m}_1 + \dot{m}_2 - \dot{m}_3)(\Delta t)$$

$$m_2 = m_1 + (\dot{m}_1 + \dot{m}_2 - \dot{m}_3)(\Delta t)$$

Inserting values

$$m_2 = 2000 \text{ lb} + \left(0.8 \frac{\text{lb}}{\text{s}} + 1.2 \frac{\text{lb}}{\text{s}} - 2.5 \frac{\text{lb}}{\text{s}}\right)(30 \text{ min}) \left| \frac{60 \text{ s}}{1 \text{ min}} \right| = 1100 \text{ lb} \quad \longleftarrow$$

Problem 4.12

As shown in Fig. P4.17, air with a volumetric flow rate of $15,000 \text{ ft}^3/\text{min}$ enters an air-handling unit at 35°F , 1 atm . The air-handling unit delivers air at 80°F , 1 atm to a duct system with three branches consisting of two 26-in.-diameter ducts and one 50-in. duct. The velocity in each 26-in. duct is 10 ft/s . Assuming ideal gas behavior for the air, determine at steady state

- (a) the mass flow rate of air entering the air-handling unit, in lb/s .
- (b) the volumetric flow rate in each 26-in. duct, in ft^3/min .
- (c) the velocity in the 50-in. duct, in ft/s .

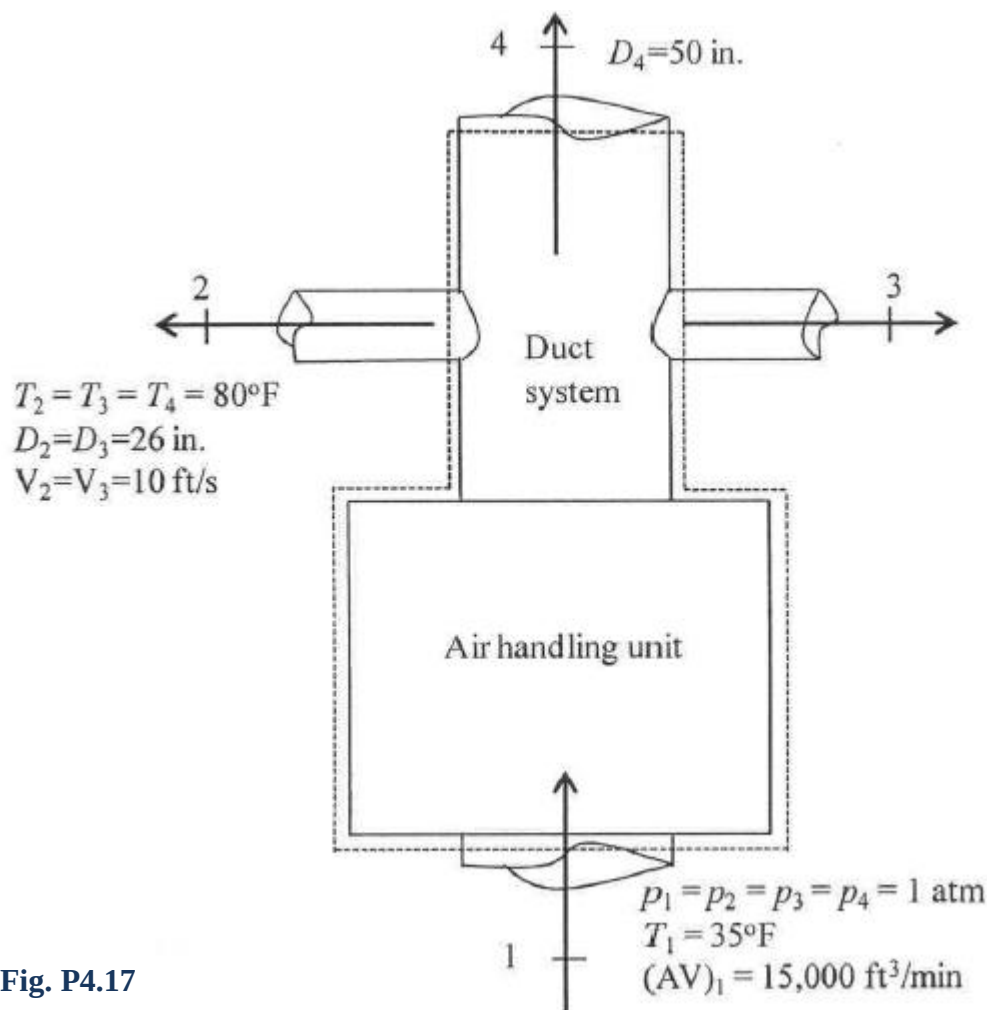


Fig. P4.17

KNOWN: At steady state, air enters an air-handling unit and exits from three ducts in an attached duct system.

FIND: Determine the mass flow rate of air entering the air handling unit, in lb/s ; the volumetric flow rate in each 26-in. duct, in ft^3/min ; and velocity in the 50-in. duct, in ft/s .

SCHEMATIC AND GIVEN DATA: See Fig. P4.17.

ENGINEERING MODEL:

- (1) The control volume shown on the schematic is at steady state.
- (2) The air behaves as an ideal gas.

Problem 4.12 (Continued)

ANALYSIS:

(a) Determine the mass flow rate of air entering the air handling unit, in lb/s, as follows:

$$\dot{m}_1 = \frac{(AV)_1}{v_1} = \frac{(AV)_1 p_1}{RT_1} = \frac{\left(15000 \frac{\text{ft}^3}{\text{min}}\right)(1 \text{ atm}) \left| \frac{14.7 \frac{\text{lbf}}{\text{in}^2}}{1 \text{ atm}} \right| \frac{144 \text{ in}^2}{1 \text{ ft}^2} \left| \frac{1 \text{ min}}{60 \text{ s}} \right|}{\left(\frac{1545 \text{ ft} \cdot \text{lbf}}{28.97 \text{ lb} \cdot \text{R}} \right)(495^\circ \text{R})} = 20.05 \frac{\text{lb}}{\text{s}} \quad \leftarrow$$

(b) Determine the volumetric flow rate in each 26-in. duct, in ft³/min, as follows:

$$(AV)_2 = (AV)_3 = \left(\frac{\pi D_2^2}{4} \right) V_2 = \left(\frac{\pi (26 \text{ in.})^2}{4} \right) \left(10 \frac{\text{ft}}{\text{s}} \right) \left| \frac{1 \text{ ft}^2}{144 \text{ in}^2} \right| \frac{60 \text{ s}}{1 \text{ min}} = 2212.2 \frac{\text{ft}^3}{\text{min}} \quad \leftarrow$$

(c) Determine the velocity in the 50-in. duct, in ft/s, as follows:

$$\frac{dm_{cv}}{dt} = 0 = \dot{m}_1 - \dot{m}_2 - \dot{m}_3 - \dot{m}_4$$

$$\dot{m}_2 = \dot{m}_3 = \frac{(AV)_2}{v_2} = \frac{(AV)_2 p_2}{RT_2} = \frac{\left(2212.2 \frac{\text{ft}^3}{\text{min}}\right)(1 \text{ atm}) \left| \frac{14.7 \frac{\text{lbf}}{\text{in}^2}}{1 \text{ atm}} \right| \frac{144 \text{ in}^2}{1 \text{ ft}^2} \left| \frac{1 \text{ min}}{60 \text{ s}} \right|}{\left(\frac{1545 \text{ ft} \cdot \text{lbf}}{28.97 \text{ lb} \cdot \text{R}} \right)(540^\circ \text{R})} = 2.71 \frac{\text{lb}}{\text{s}}$$

$$\dot{m}_4 = \dot{m}_1 - \dot{m}_2 - \dot{m}_3 = 20.05 \frac{\text{lb}}{\text{s}} - 2 \left(2.71 \frac{\text{lb}}{\text{s}} \right) = 14.63 \frac{\text{lb}}{\text{s}}$$

$$\dot{m}_4 = \frac{(AV)_4}{v_4}$$

$$V_4 = \frac{\dot{m}_4 v_4}{A_4} = \frac{\dot{m}_4 RT_4}{p_4 \left(\frac{\pi D_4^2}{4} \right)} = \frac{\left(14.63 \frac{\text{lb}}{\text{s}}\right) \left(\frac{1545 \text{ ft} \cdot \text{lbf}}{28.97 \text{ lb} \cdot \text{R}} \right)(540^\circ \text{R})}{(1 \text{ atm}) \left(\frac{\pi (50 \text{ in.})^2}{4} \right) \left| \frac{1 \text{ atm}}{14.7 \frac{\text{lbf}}{\text{in}^2}} \right|} = 14.6 \frac{\text{ft}}{\text{s}} \quad \leftarrow$$

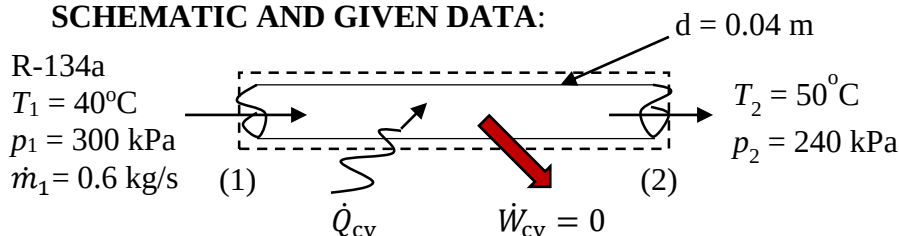
Problem 4.16

Refrigerant 134a enters a horizontal pipe operating at steady state at 40°C, 300 kPa, and a mass flow rate of 0.6 kg/s. At the exit, the temperature is 50°C and the pressure is 240 kPa. The pipe diameter is 0.04 m. Determine (a) the velocities at the inlet and exit, in m/s, and (b) the rate of heat transfer between the pipe and its surroundings, in kW.

KNOWN: Data are provided for Refrigerant 134a flowing through a horizontal, constant-area pipe at steady state.

FIND: Determine the velocities at the inlet and exit and the rate of heat transfer between the pipe and its surroundings.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL: (1) The control volume shown in the sketch is at steady state. (2) For the control volume, $\dot{W}_{cv} = 0$, and $\Delta pe = 0$ (horizontal).

ANALYSIS: (a) The inlet velocity is found using the expression for mass flow rate: $\dot{m}_1 = \frac{A_1 V_1}{v_1}$.

$$V_1 = \frac{\dot{m}_1 v_1}{A_1}$$

The area is $A_1 = \pi d^2/4 = \pi (0.04 \text{ m})^2/4 = 1.257 \times 10^{-3} \text{ m}^2$. The specific volume v_1 is obtained from Table A-12 at 40°C, 300 kPa = 3 bar by interpolation: $v_1 \approx 0.08089 \text{ m}^3/\text{kg}$. Thus

$$V_1 = \frac{(0.6 \text{ kg/s})(0.08089 \frac{\text{m}^3}{\text{kg}})}{(1.257 \times 10^{-3} \text{ m}^2)} = 38.6 \text{ m/s}$$

To get the exit velocity, apply the mass rate balance at steady state to get $\dot{m}_1 = \dot{m}_2 \equiv \dot{m}$. For the constant diameter pipe

$$\frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} \rightarrow V_2 = \frac{v_2}{v_1} V_1$$

State 2 is also in the superheated vapor region, so from Table A-12: $v_2 = 0.10562 \text{ m}^3/\text{kg}$, and

$$V_2 = \frac{(0.10562)}{(0.08089)} (38.6 \text{ m/s}) = 50.4 \text{ m/s}$$

(b) To find the rate of heat transfer, we apply the energy balance for the one-inlet, one-exit control volume at steady state

Problem 4.16 (Continued)

$$\cancel{\frac{dE_{cv}}{dt}} = \dot{Q}_{cv} - \cancel{\dot{W}_{cv}} + \dot{m}[(h_1 - h_2) + (V_1^2 - V_2^2)/2 + \cancel{g(z_1 - z_2)}]$$

Or

$$\dot{Q}_{cv} = \dot{m}[(h_2 - h_1) + (V_2^2 - V_1^2)/2]$$

With data for specific enthalpies from Table A-12

$$\begin{aligned}\dot{Q}_{cv} &= (0.6 \text{ kg/s})[(294.47 - 284.05) \frac{\text{kJ}}{\text{kg}} + \left[\frac{50.4^2 - 38.6^2}{2} \right] \left(\frac{\text{m}}{\text{s}} \right)^2 \left| \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right| \left| \frac{1 \text{ kJ}}{10^3 \text{ N} \cdot \text{m}} \right| \\ &= 6.567 \text{ kJ/s} \left| \frac{1 \text{ kW}}{1 \text{ kJ/s}} \right| = 6.567 \text{ kW (in)} \quad \longleftarrow\end{aligned}$$

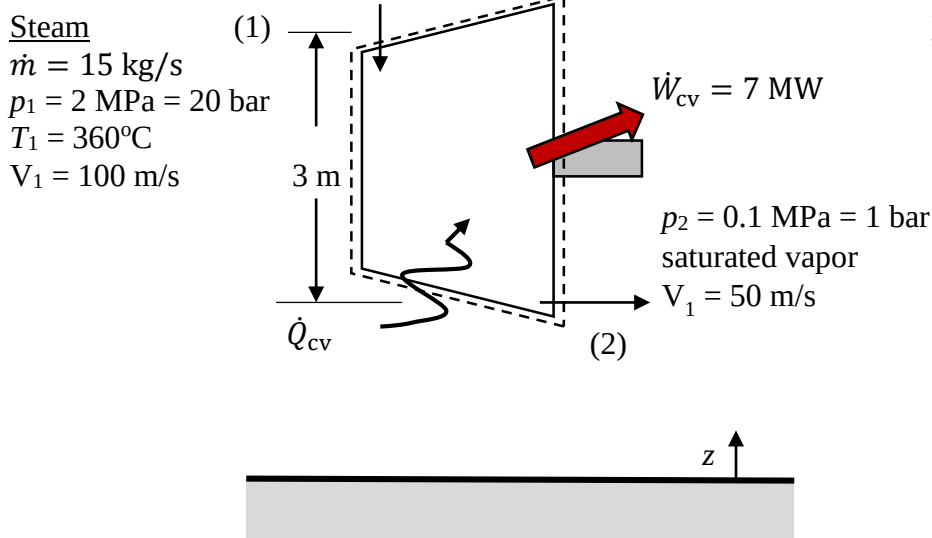
Problem 4.32

Steam enters a turbine operating at steady state at 2 MPa, 360°C with a velocity of 100 m/s. Saturated vapor exits at 0.1 MPa and a velocity of 50 m/s. The elevation of the inlet is 3 m higher than at the exit. The mass flow rate of steam is 15 kg/s, and the power developed is 7 MW. Let $g = 9.81 \text{ m/s}^2$. Determine (a) the area at the inlet, in m^2 , and (b) the rate of heat transfer between the turbine and its surroundings, in kW.

KNOWN: Data are provided for the steady-state operation of a steam turbine. The power and mass flow rate are specified.

FIND: Determine the inlet area and the rate of heat transfer between the turbine and its surroundings.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL: (1) The control volume shown on the sketch is at steady state.
(2) $g = 9.81 \text{ m/s}^2$

ANALYSIS: (a) The mass rate balance for the one-inlet, one-exit control volume at steady state reduces to: $\dot{m}_1 = \dot{m}_2 \equiv \dot{m}$. Thus

$$\dot{m}_1 = \frac{A_1 V_1}{v_1} \rightarrow A_1 = \frac{\dot{m} v_1}{V_1}$$

State 1 is in the superheated vapor region, so from Table A-4: $v_1 = 0.1411 \text{ m}^3/\text{kg}$ and

$$A_1 = \frac{(15 \frac{\text{kg}}{\text{s}})(0.1411 \frac{\text{m}^3}{\text{kg}})}{(100 \frac{\text{m}}{\text{s}})} = 0.021 \text{ m}^2$$

(b) Reducing the energy balance at steady state for the one-inlet, one-exit control volume (Eq. 4.20a)

$$0 = \dot{Q}_{\text{cv}} - \dot{W}_{\text{cv}} + \dot{m}[(h_1 - h_2) + (V_1^2 - V_2^2)/2 + g(z_1 - z_2)]$$

Solving

Problem 4.32 (Continued)

$$\dot{Q}_{cv} = \dot{W}_{cv} + \dot{m}[(h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(z_2 - z_1)] \quad (1)$$

Evaluating (1) term-by-term with data from Table A-3 for State 1 and Table A-4 for State 2:

$$h_2 - h_1 = (2675.5 - 3159.3) \text{ kJ/kg} = -483.8 \text{ kJ/kg}$$

$$(V_2^2 - V_1^2)/2 = \left[\frac{(50 \frac{\text{m}}{\text{s}})^2 - (100 \frac{\text{m}}{\text{s}})^2}{2} \right] \left| \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right| \left| \frac{1 \text{ kJ}}{10^3 \text{ N} \cdot \text{m}} \right| = -3.75 \text{ kJ/kg}$$

① $g(z_2 - z_1) = (9.81 \text{ m/s}^2)(3 \text{ m}) \left| \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right| \left| \frac{1 \text{ kJ}}{10^3 \text{ N} \cdot \text{m}} \right| = -0.03 \text{ kJ/kg}$

Inserting values in (1)

$$\begin{aligned} \dot{Q}_{cv} &= (7 \text{ MW}) \left| \frac{10^3 \text{ kW}}{1 \text{ MW}} \right| + (15 \text{ kg/s}) [(-483.8) + (-3.75) + (-0.03)] \text{ kJ/kg} \left| \frac{1 \text{ kW}}{1 \text{ kJ/s}} \right| \\ &= -313.7 \text{ k W (out)} \end{aligned} \quad \longleftarrow$$

1. Note that kinetic and potential energy effects are relatively minor.